

Derivative:

$$g(x) = \sqrt{\frac{3 - \sin x}{1 + \cos(2x)}}$$

$$W = \frac{3 - \sin x}{1 + \cos(2x)} \Rightarrow \begin{aligned} u &= 3 - \sin x & u' &= -\cos x \\ v &= 1 + \cos(2x) & v' &= 2(\sin(2x)) \end{aligned}$$

$$W' = \frac{-(3 - \sin x)(2(\sin(2x))) + (1 + \cos(2x))(-\cos x)}{(1 + \cos(2x))^3}$$

$$g'(x) = \frac{\frac{(-\cos x)(1 + \cos(2x)) - (3 - \sin x)(2\sin(2x))}{(1 + \cos(2x))^2}}{2\sqrt{\frac{3 - \sin x}{1 + \cos(2x)}}}$$

Simplify:

$$\text{Let } L = (-\cos x)(1 + \cos(2x)) - (3 - \sin x)(2 \sin(2x))$$

$$L = -\cos x - \cos x \cdot \cos(2x) - 6 \sin(2x) + 2 \sin x + 4 \sin^2 x \cos x$$

Note: $\sin x \cdot 2 \sin(2x)$
 $\rightarrow \sin(2x) = 2 \sin x \cdot \cos x$
 $\Rightarrow (\sin x)(2 \sin x)(\cos x)(2) = 4 \sin^2 x \cdot \cos x$

$$L = -\cos x - \cos x \cdot 2 \cos^2 x - 1 - 6(2 \sin^2 x \cos x) + 2 \sin x + 4 \sin^2 x \cos x$$

$$L = -\cos x - \cos x \cdot 2 \cos^2 x - 1 - \underline{12 \sin^2 x \cos x} + 2 \sin x + \underline{4 \sin^2 x \cos x}$$

$$L = -\cos x - \cos x \cdot 2\cos^2 x - 1 - 8\sin^2 x \cos x + 2\sin x$$

$$g'(x) = \frac{-\cos x - \cos x \cdot 2\cos^2 x - 1 - 8\sin^2 x \cos x + 2\sin x}{(1 + \cos(2x))^2}$$

$$2 \sqrt{\frac{3 - \sin x}{1 + \cos(2x)}}$$